

MA-SQLGrid: A Robust Multi-Agent Framework for Text-to-SQL in Power Grid Databases

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Featured Application

MA-SQLGrid is an experimental read-only decision-support framework for auditable natural-language access to structured power-grid records. Generated SQL and returned source rows require human inspection before consequential use; no operational control, protection, or maintenance decision has been validated.

Abstract

Natural-language access to power-grid databases requires traceable schema grounding, database-enforced read-only execution, and explicit handling of missing evidence. We operationalize “robust” only as tested mutation denial, bounded execution, complete-evidence gating, and constructed-witness behavior; universal or deployment robustness is not claimed. We present MA-SQLGrid, a five-role software architecture coordinated through an append-only blackboard and deterministic adjudicator. Its implemented boundary combines SQLite read-only mode, query_only, an authorizer, disabled extensions, resource limits, and fail-closed named-state coverage. Earlier GridDB and BIRD experiments evaluate frozen prompting workflows rather than autonomous multi-agent generation. A separate 700-call study uses a prospectively frozen call-and-selection procedure on development-visible GridDB items to test presented-value and reference-free selection components. A no-generation release-v3 re-execution applies three selectors to a shared 5760-attempt collection for eight historical candidates on 180 GridDB questions. Because frozen pre-run tests accessed same-item gold-derived outcomes, this release is descriptive. Fixed order, validation-only ranking, and complete three-witness selection obtained 80/180, 100/180, and 101/180. The two adjudicating selectors tied at the highest score on 130/180 questions; reversing candidate order produced 117–118/180. Their sole difference, Q039, is a constructed and semantically ambiguous projection trace. The evidence supports a reproducible architecture with tested read-only execution and bounded-evidence mechanisms, but not five-role superiority, qualified power-grid semantic validity, deployment safety, or universal robustness.

Keywords: text-to-SQL; multi-agent systems; power-grid databases; schema grounding; execution validation; constructed-state testing; reproducibility

Received:

Accepted:

Published:

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1. Introduction

Power-grid organizations maintain relational data about assets, locations, topology, sensor readings, work orders, technicians, and maintenance activity. SQL provides precise

access to these records, but it requires knowledge of table names, join keys, identifiers, units, aggregation conventions, and local coding rules. Text-to-SQL systems aim to reduce this access barrier by translating a natural-language request into an executable query. In an engineering workflow, however, a fluent query is not sufficient evidence of correctness. A query can execute while selecting the wrong status field, omitting a required filter, using an unintended join path, or returning the expected number of columns with the wrong content.

The risk is amplified by the semantics of grid records. An identifier that looks numeric may be a categorical code; a timestamp can refer to sensing, issuance, acknowledgement, or closure; and an asset status may be current, event-time, or work-order-specific. The same user wording can therefore map to several executable queries. Natural-language access must expose schema and value grounding, preserve the generated statement, and retain the returned source rows so that an engineer can inspect the interpretation. It should also abstain when the available evidence cannot distinguish candidates.

Cross-database benchmarks such as Spider and BIRD have made schema generalization and realistic database content central evaluation concerns [1,2]. Spider 2.0 extends this trajectory to enterprise workflows with large schemas and multiple SQL dialects, while MultiSQL broadens context-dependent evaluation across diverse SQL operations [3,4]. Schema-aware models, constrained decoding, context selection, decomposition, and execution guidance address different failure surfaces [5–9]. These methods also expose a design problem: generation steps are frequently bundled together. A system may change schema serialization, value presentation, domain normalization, structural hints, candidate count, and repair policy simultaneously. An observed score difference then cannot be assigned cleanly to schema linking, reasoning, or validation.

Recent Applied Sciences studies illustrate the breadth of this design space. RGISQL refines grammatical information in a relational-graph model, while zero-shot prompting work compares prompt strategies and benchmark behavior [10,11]. Schema-retrieval systems use embeddings and vector stores to limit context, and domain-specific work such as DKA-SQL adapts knowledge to power-grid questions [12,13]. These contributions focus primarily on generation quality or contextualization. The present work instead asks what can be established when candidate production, validation evidence, state-based diagnostics, and deterministic selection are recorded as separate stages.

The original MA-SQLGrid concept addresses this problem through five specialized roles. A Query Analyst interprets the request, a Schema Cartographer maps it to the database, an SQL Synthesizer packages candidate queries, a Validation Engine checks safety and execution, and a role historically named the Counterfactual Critic—operationally a Metamorphic-State Critic—records evidence from named constructed database states. A separate deterministic adjudicator resolves the resulting evidence. This separation makes the trace inspectable: a reviewer can distinguish what the analyst inferred, which schema elements were retained, which candidates were proposed, which candidates executed, and why a final candidate was selected or rejected.

The architecture must be separated from its evidence. GridDB used one generation per question and prompt cell followed by deterministic validation and scoring; BIRD compared four frozen prompting procedures. Neither executed five interacting agents. Likewise, selecting from eight historical candidates can recover an existing correct query without improving generation, and the mixed-model pool has a larger physical-call history than a one-shot baseline. We therefore separate generation coverage, safe executability, reference-free ranking, and gold-relative accuracy. The contribution is an evidence architecture in which a future matched experiment can replace one role while retaining message schemas, executor, state requirements, call accounting, and the gold boundary.

This distinction leads to three questions. First, does the implemented five-role software conform to its typed handoff, database-enforced, gold-isolated, and fail-closed contracts? This is a software-conformance question, not an efficacy comparison. Second, what do the completed experiments establish about context packages, presented values, public-database transfer, and constructed-state agreement? Third, when generation is held fixed at eight already frozen candidates per question, how do fixed-order, validation-only, and complete-three-witness selectors behave under the same evidence-collection operations? The third question estimates historical-pool offline selection only; schema grounding is retained as trace evidence and no five-role end-to-end, new-generation, dialogue, or deployment effect is estimated.

The analysis uses an evidence hierarchy: software tests address mutation denial, resource termination, trace completeness, and fail-closed coverage; frozen experiments address prompt and model behavior; BIRD addresses non-grid portability; and release v3 addresses deterministic selection from a historical pool with prior same-item outcome access disclosed. This prevents software conformance from being presented as semantic correctness or descriptive selection as prospective multi-agent superiority.

The paper makes four contributions. First, it defines and tests software conformance for a five-role coordination core, a database-enforced SQLite executor, and a deterministic adjudicator whose required counterfactual coverage fails closed; no experiment estimates a five-role benefit. Second, it conservatively integrates a 1440-prediction GridDB factorial experiment, an audited 700-call component study with a prospectively frozen procedure on development-visible items, a 25,920-row multi-state ledger, and a 5000-call BIRD comparison while retaining their original identities and evidence classes. Third, it reports a deterministic descriptive re-execution over the existing eight-slot pool using three query-blind constructed witnesses with distinct database hashes; all methods choose only after the same shared 32-attempt evidence collection per question is complete. Fourth, it publishes complete counts and adverse diagnostics: 130/180 top-score ties for both adjudicating selectors, approximately 5.4 candidates per top-tie set, all 18 order/weight/threshold cells, and the semantically ambiguous 1/180 Q039 trace.

No Spider or WikiSQL result or unsupported large grid question-SQL corpus is asserted. In the retained title, *multi-agent* names the typed software decomposition and *robust* only the tested mutation-denial, bounded-execution, complete-evidence, and three-witness mechanisms. Qualified domain review and a call-matched new-generation experiment remain mandatory before broader interpretation.

2. Related Work

2.1. Text-to-SQL Evaluation and Schema Generalization

Text-to-SQL systems map a natural-language question and a database schema to a query. Spider established a cross-domain setting with previously unseen databases, drawing attention to table selection, joins, grouping, nesting, and ordering [1]. BIRD extends public evaluation toward larger and more content-rich databases [2]. A benchmark result depends on the exact split, evaluator, database engine, execution boundary, and model configuration. We therefore report only the completed BIRD Mini-Dev protocol and make no claim about public datasets that were discussed but not run.

Exact SQL match and execution evaluation answer different questions. String match can penalize equivalent SQL, whereas result equality can reward nonequivalent queries on an insufficiently discriminating database state. A robustness evaluation should retain predicted SQL, record every failure, and specify the state on which equality was observed. MA-SQLGrid uses strict result equality for inherited snapshot analyses and separately re-

ports constructed-state agreement and bounded metamorphic invariance. Neither auxiliary endpoint is semantic certification.

Execution equivalence is particularly vulnerable when a table is small, a predicate happens not to filter any row, or two joins coincide on one snapshot. Test-suite accuracy and multiple database states can reduce this accidental agreement, but their value depends on how states are constructed and whether their transformations preserve the intended query semantics. Our prediction-blinded state suite is therefore reported as a constructed diagnostic. The three release-v3 witnesses are even narrower: they test invariance to irrelevant relations, storage/index changes, and nullable extensions, not correctness under realistic grid evolution.

Work on SQL-oriented language-model teams shows another distinction between using several models as options and coordinating explicit roles [14]. Recent Text-to-SQL systems operationalize multi-agent collaboration through decomposition and tool use, question-dependent agent routing, or review–rebuttal–revision consensus [15–17]. A team can increase candidate diversity, whereas an agent architecture can formalize ownership of intent analysis, schema grounding, generation, validation, and critique. Either may help, but a fair benefit claim requires equal model calls, candidate counts, and selection information. The current paper supplies the role contracts and a fixed-pool diagnostic, not that matched end-to-end comparison.

2.2. Schema Linking, Context Packages, and Output Contracts

Schema linking connects question spans to tables, columns, values, and foreign-key relations [5,18,19]. Full-schema context preserves recall but can increase token cost and distraction. Compact context reduces the search space but risks dropping a required join key or predicate. Value examples can disambiguate coded attributes, while question-derived instructions can identify aggregation, result shape, ordering, or a limit.

Retrieval-based schema selection frames this trade-off as a recall problem over relevant tables, columns, and relationships [12]. Recent systems strengthen this stage through robust in-context schema linking, generation-driven multi-model learning, and scalable multi-database retrieval [20–22]. Graph pathfinding and bidirectional table–column retrieval provide complementary mechanisms for retaining join structure while reducing irrelevant context [23,24]. In a grid database, lexical relevance alone may miss bridge tables or coded fields whose names do not occur in the question. A safe compact package should retain foreign-key closure, report unmatched question spans, and abstain when required relations cannot be justified. The current Cartographer records these elements, but its lexical rules are an inspectable baseline rather than a learned retrieval contribution.

The inherited GridDB factorial study isolates two implemented prompt factors: a full-DDL/global-values package versus a compact/domain-grounded package, and absence versus presence of a composite structural/SQL-operation hint. The compact package is not a pure schema-length manipulation because it also changes value presentation and adds corpus-tailored normalization. The hint is not merely formatting because it can include aggregation, grouping, projection, ordering, and limit guidance. We retain these precise labels instead of assigning broader causal interpretations.

Constrained generation and execution guidance reduce invalid candidates. PICARD constrains decoding so inadmissible SQL continuations can be rejected during generation [6]. CHESS combines contextual grounding and SQL synthesis components [9]; other prompt-based methods investigate decomposition and representation choices [7,8,25]. Recent workflow methods add question rewriting with execution-guided refinement, attention-oriented decomposition, or partition-and-drill reasoning [26–28]. Execution feedback is also used for preference optimization, active database probing, and dialect-

aware bootstrapping [29–31]. Synthetic strong- and weak-model data offer a separate route to specialized open-source Text-to-SQL models [32]. MA-SQLGrid differs at the coordination boundary: its new core accepts externally produced candidates, checks them with deterministic contracts, and records evidence before selection. It does not claim that the current lexical cartographer supersedes learned schema-linking systems.

This boundary also changes the reproducibility unit. A monolithic prompt often records only the final SQL, while the blackboard records the question-only intent, schema links, canonical candidates, safety decisions, execution hashes, named-state coverage, and final rule outcome. Intermediate records do not guarantee correct reasoning, but they allow an audit to locate whether a failure arose before generation, at parsing, under database authorization, during evidence collection, or at adjudication.

2.3. Agentic Decomposition and Auditable Control

Agentic decomposition exposes intermediate decisions, but benefit requires holding model, data, decoding, candidate count, and physical generation budget constant. Energy-management agents represent physical or decision entities [33]; MA-SQLGrid instead assigns typed software responsibilities over an information trace and controls no grid equipment. Roles may be deterministic or model-assisted: the Synthesizer can package frozen candidates, the Validator applies fixed policy, the Critic consumes named-state evidence, and adjudication is deterministic. Append-only traces record calls, retries, and visibility; the current no-call offline study tests these interfaces without estimating conversational benefit.

2.4. Power-Grid Data and External Validity

Power-system datasets combine topology, equipment attributes, measurements, time series, costs, and constraints. RTS-GMLC and SimBench provide reproducible engineering resources that can be transformed into relational structures [34,35]. Their availability does not automatically create a text-to-SQL benchmark: intended projections, units, ordering, ties, and reference SQL require semantic review. The current RTS-GMLC, SimBench, and NERC-derived candidates are consequently machine-adjudicated silver resources, not expert gold.

DKA-SQL is a direct power-grid text-to-SQL precedent [13]. Its published corpus, method, models, and evaluation boundary differ from ours. Because no official implementation was used, the current BIRD protocol is neither a DKA-SQL reproduction nor a head-to-head comparison.

External validity has two axes. Database structure must resemble target assets, topology, measurements, and maintenance records, while questions and reference answers must reflect how qualified users formulate and resolve information needs. RTS-GMLC and SimBench contribute the first axis but not automatically the second. BIRD contributes a public language-to-database benchmark but is not grid-specific. GridDB supplies a controlled conjunction of both axes, yet remains synthetic, development-visible, and without completed independent expert adjudication.

Table 1 summarizes the closest methodological axes. The comparison is architectural rather than a score table because datasets, call budgets, and evaluators differ. The contribution lies in combining typed role traces with database-enforced execution and complete named-state eligibility; prospective evaluation on expert-reviewed power-grid queries remains necessary.

Table 1. Positioning against adjacent Text-to-SQL approaches; entries describe reported design emphasis, not comparable performance.

Approach	Principal emphasis	Evaluation setting	Difference from MA-SQLGrid
RGISQL [10]	Grammatical and relational graph modeling	Public Text-to-SQL tasks	Learned generator rather than a traced executor/adjudicator boundary
Zero-shot prompting [11]	Prompt strategies and LLM behavior	Public benchmarks	Prompt comparison without the five-role blackboard
Schema retrieval [12]	Embedding/vector-store context selection	LLM SQL generation	Retrieval quality rather than fail-closed named-state selection
DKA-SQL [13]	Domain knowledge adaptation	Power-grid Text-to-SQL	Different corpus and implementation; not reproduced here
MA-SQLGrid	Typed trace, read-only executor, deterministic selection	GridDB case study plus BIRD portability	Prospective five-role benefit and expert semantic validation remain open

3. Materials and Methods

3.1. Task and Evidence Boundary

For question x_i , read-only database D_i , and introspected schema S_i , a generator produces one or more candidates

$$\mathcal{Y}_i = \{\tilde{y}_{i1}, \dots, \tilde{y}_{ik}\}. \tag{1}$$

Each candidate must satisfy a single-statement read-only contract. The accepted leading form is SELECT or WITH; comments, multiple statements, write operations, schema changes, attachments, and pragmas are rejected. A candidate is eligible for adjudication only when it is safe and executable.

The offline evaluator may later compare the selected query with a reference result. Let $E_i = 1$ when the selected SQL executes and its result equals the frozen reference result under the registered evaluator, and $E_i = 0$ otherwise. Reference SQL, reference results, E_i , and any derivative correctness field are forbidden from all five agents and the Adjudicator. Evaluation occurs only after the blackboard is sealed.

This boundary yields three distinct objects. A candidate is *admissible* when its syntax and requested operations satisfy policy, *executable* when it completes within the registered database limits, and *correct under the frozen evaluator* when its result matches the reference after sealing. Admissibility is neither executability nor correctness. The manuscript preserves all three labels because collapsing them would let an executable but semantically wrong query appear validated.

The framework’s response contract contains the selected SQL, stable candidate and source identifiers, referenced tables and columns, execution-state and result hashes, state-coverage summary, abstention or selection reason, and the sealed trace digest. It does not present generated text as an authorization decision. A calling application must separately authenticate the user, restrict the database view, and decide whether displaying the returned rows is lawful. Those controls lie outside the evaluated framework.

3.2. Data Resources and Provenance

GridDB-Maintenance-v2 v0.1 is a synthetic SQLite case study with eight tables and 98 rows: `asset_types` (6), `assets` (18), `grid_topology` (9), `locations` (8), `maintenance_logs` (8), `sensor_readings` (26), `technicians` (8), and `work_orders` (15). It contains 200 natural-

language–SQL records: 20 development records and 180 factorial evaluation records. The evaluation records include 66 easy, 91 medium, and 23 hard items and map to 70 normalized gold–SQL structural clusters, of which 58 are singletons. The evaluation partition had been visible during project development; it is a controlled finite-set case study, not a sealed production benchmark.

BIRD Mini-Dev contributes 500 public items over 11 SQLite databases. The protocol froze order, prompts, two local models, evaluator, Python 3.10.11/SQLite 3.40.1, and zero retries. Each backbone made 2500 calls and produced 2000 final predictions; independent re-execution found zero score mismatches. Two earlier failed attempts totaling 2476 calls remain immutable excluded incidents.

The GridDB reliability release contains 18 states: the canonical snapshot, three insertion permutations, and 14 additional schema-valid stress states. The scorer produced 25,920 prediction-state rows (1440×18). A prediction-blinded automated protocol admitted a 66-question subset for the primary 15-state analysis and retained 114 order-sensitive questions as diagnostics. These constructed states are not operator-certified grid snapshots.

Artifacts are labeled Inherited, Recomputed, New, or Diagnostic. The coordination core is implementation evidence and its replay is Diagnostic. Neither is a New execution-accuracy result.

Evidence classes preserve chronology: Inherited predictions and outcomes predate the framework; Recomputed processes retained inputs without new generation; New requires inputs, rules, and visibility to be frozen before outcome access; and Diagnostic permits post hoc inspection only. Labels apply per result because the manuscript contains several classes.

Table 2 separates resources by domain, evidence class, visibility, and semantic-review status. BIRD is explicitly non-grid. RTS-GMLC and SimBench provide authentic power-system structures, but their question–SQL pairs remain machine-generated silver candidates with zero completed qualified-human reviews and zero sealed items.

Table 2. Data-resource scope and evidence status. Counts describe retained local artifacts; they are not interchangeable benchmark denominators.

Resource	DBs	Tables	Rows	NL–SQL	Evidence class	Visibility and review status
GridDB v0.1	1	8	98	200	Synthetic grid; inherited development case	20 dev + 180 evaluation; development-visible; no documented independent dual-expert review
GridDB state suite	18 states	8 state-dependent	0 new	0	Constructed grid-state diagnostic	Prediction-blinded state construction; not operator-certified
RTS-GMLC pilot	1	10	360,530	55	Authentic grid structure; machine silver	AUTO_CANDIDATE; 0 human-reviewed; 0 sealed; source-use notice applies
SimBench pilot	1	8	874	36	Authentic grid structure; machine silver	Development-visible; 0 human-reviewed; 0 sealed; ODbL/DbCL review required before redistribution
BIRD Mini-Dev	11 dataset-specific	dataset-specific	500	Public portability benchmark	non-grid	Frozen public split; BIRD v1.1 protocol; not evidence of power-grid validity

3.3. Five-Agent Coordination Framework

Table 3 gives the master protocol-to-estimand map used throughout Results. It prevents denominators, visibility histories, and inferential licenses from being pooled across studies.

Table 3. Master protocol, estimand, visibility, and claim-license map. “Composition interval” denotes resampling sensitivity of the observed finite corpus, not a population confidence interval.

Protocol	Unit and <i>N</i>	Dependence proxy	Generation budget	Endpoint	Gold visibility	Status/multiplicity	Strongest licensed claim
GridDB factorial	question; 180/cell	70 normalized-SQL groups	1 call/item/cell	strict execution equality	development-visible items; scored after generation procedure frozen before calls; items development-visible	finite-set; 9-test Holm family	prompt-package behavior, not five-role benefit
Component E1/E2	question; 170 paired or 180	61 or 70 groups	700 total calls	execution equality	development-visible items; scored after generation procedure frozen before calls; items development-visible	composition intervals; registered families	component behavior only
Formal-v5 states	question; 66 primary	12 structural groups	0 new calls	15-state AND	state construction prediction-blind; gold-relative scoring later	exact 2 ¹² signs; 9-test Holm	constructed-state agreement
BIRD v1.1	item; 500 across 11 databases	database	B0-B2: 1; B3: 2 calls/item	execution accuracy	public benchmark; evaluated after calls	12 pairwise tests with Holm	non-grid workflow portability under unequal calls
Replay	question; 180	none	0 new calls	pool coverage	prior evidence existed	descriptive	interface feasibility only
Release v3	question; 180	none	shared 5760 evidence attempts	historical-pool execution match	same-item derived outcomes previously accessed	outcome-exposed descriptive sensitivity	deterministic selector behavior, not coordination efficacy

Table 3 also records the decisive chronology. GridDB factorial outcomes are inherited and development-visible; the component call-and-selection procedure was frozen before calls on development-visible items; formal-v5 states were constructed prediction-blind; BIRD used a frozen public protocol; and replay used existing predictions without accuracy scoring. Release v3 sealed decisions before directly loading raw gold, but its regression suite had previously read same-item v2 outcomes. Hashing and within-run sealing therefore provide traceability, not prospective outcome blindness.

MA-SQLGrid comprises five specialist roles coordinated by a deterministic controller. Figure 1 separates the five typed roles, append-only evidence trace, read-only execution boundary, deterministic adjudication, and post-sealing evaluation boundary. The controller is not a sixth reasoning model: it orders calls, applies the registered rule, and seals the trace. Messages receive consecutive sequence numbers, and the complete trace is serialized canonically to a SHA-256 digest.

The **Query Analyst** produces a structured question-only intent record containing lexical tokens, detected aggregations, ordering requirements, and an explicit limit when present. It does not select SQL or see correctness. The **Schema Cartographer** maps lexical intent to tables, columns, and foreign-key edges. The current skeleton uses deterministic overlap and records unmatched tokens; a richer GridDB adapter adds disclosed domain rules and graph expansion.

The **SQL Synthesizer** packages external SQL without model calls, canonicalizes formatting, removes exact duplicates while preserving first occurrence, and assigns stable identifiers. The release-v3 **Validation Engine** opens a frozen database using `mode=ro&immutable=1`, enables `query_only`, disables extensions, and installs an authorizer. It denies writes, DDL, transactions, attachments, pragmas, metadata enumeration, dangerous functions, and reads outside table/column allowlists; progress and fetch handlers enforce time, opcode, and row limits. Every attempt retains database/SQL hashes, failures, timing, row count, and result hash without hidden retry.

The later FINAL executor adds function allowlisting, projected-width, raw-cell-byte, and total-result-byte limits. Fourteen tests cover 1/5/50 MB BLOB denial, total-budget and exact-boundary behavior, trace retention, oversized text/results, and wide projections. These controls were not used for the v3 ledger or 80/100/101 counts. Table/column allowlists constrain query scope but do not authenticate users or provide row-level au-

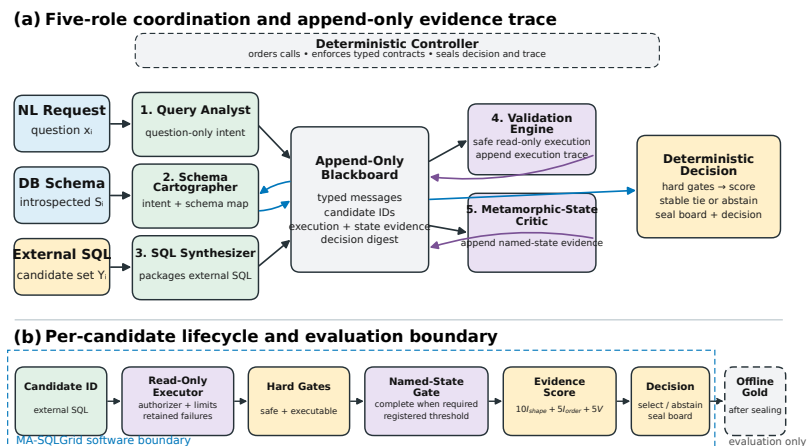


Figure 1. Implemented MA-SQLGrid information flow and candidate lifecycle. **(a)** The Query Analyst and Schema Cartographer post question and schema records to an append-only blackboard, and the SQL Synthesizer packages externally supplied candidates. The Validation Engine records read-only execution evidence, while the Metamorphic-State Critic records named-state evidence when required. **(b)** Hard eligibility gates precede the effective evidence score $e(y) = 10I_{\text{shape}} + 5I_{\text{order}} + 5V$, $V \in [0, 1]$, and the deterministic tie or abstention rule. Gold or reference results are loaded only after the decision and board digest are sealed. User authorization and operational approval remain outside the software boundary.

thorization; both variants establish tested SQLite mutation denial and scoped execution bounds, not data minimization, process isolation, or operational approval.

The threat model assumes untrusted SQL but a trusted host, SQLite library, configured allowlist, and immutable database. It does not address a compromised process, OS escape, side channels, built-in malicious extensions, or upstream authorization errors. Production use requires process isolation, least-privilege views, authenticated context, logging, and independent security review.

Failure handling is intentionally fail-closed and non-retrying. A parse denial, authorizer denial, timeout, opcode limit, row limit, width limit, cell limit, or total-result limit creates a retained attempt record and makes that candidate ineligible for the current decision. The controller cannot silently weaken a limit or substitute a repaired query. A repair procedure may be studied as a separate registered generation step, as in BIRD B3, but its extra physical call and visibility must remain explicit.

The **Metamorphic-State Critic** (the historically named Counterfactual Critic) consumes named state results, rejects duplicate state identifiers, and records evaluated, passed, failed, and missing states. When state evidence is optional, missing evidence remains diagnostic and does not break prior no-state behavior. When the caller requires state evidence, coverage must equal the complete registered state set and the pass threshold must be met; otherwise the candidate is ineligible. Thus an incomplete 1/1 record cannot outrank a complete 10/11 record. Existing formal-v5 labels compare predictions with gold and remain forbidden during selection.

The blackboard is append-only at the application layer: a role posts a typed message and cannot rewrite an earlier message. The controller checks role, message type, sequence number, and required fields before acceptance. Canonical serialization fixes key order and whitespace before hashing. This makes two runs comparable at the trace level and

exposes an unexpected hidden field or changed handoff. It does not prevent a malicious host process from altering memory or files; process isolation and remote attestation are outside scope.

Information flow is intentionally asymmetric. The Analyst receives the question but no candidate SQL. The Cartographer receives the question, intent record, and introspected schema. The Synthesizer receives external candidate strings and assigns identities but cannot execute them. The Validator receives one candidate plus a scoped executor. The Critic receives named-state outcomes but no reference correctness. Only the deterministic Adjudicator receives the eligible evidence summaries. Gold enters a separate offline evaluator after the decision message and board digest exist.

3.4. Deterministic Adjudication and Abstention

Eligibility always requires safety and successful execution. The archived 100-point representation assigns 40 points for each of those two properties, followed by 10 for question-derived aggregate-shape conformity, 5 for question-derived ordering conformity, and up to 5 for lexical value hits. Because every ranked candidate has already passed both hard gates, the first 80 points are constant and contain no ranking information. The effective evidence score is therefore $e(y) = 10I_{\text{shape}}(y) + 5I_{\text{order}}(y) + 5V(y)$ for an eligible candidate y , with $V(y) \in [0, 1]$. When named-state evidence is optional, the controller selects an $\arg \max e(y)$ and ties preserve frozen candidate order. When named-state evidence is required, incomplete coverage or failure of the registered pass threshold makes y ineligible before the same ranking. If the eligible set is empty, the controller abstains. This piecewise gate–score–tie function is mathematically equivalent to the archived implementation; it exposes that safety/execution are eligibility controls rather than discriminative score evidence. V3 does not establish that the weight design preceded access to derived outcomes for the 180 items, so weight, pass-threshold, and reversed-order results remain descriptive sensitivity analyses.

The archived 40/40/10/5/5 allocation—equivalently, hard safety/execution gates followed by a 10/5/5 evidence score—is an illustrative deterministic policy, not a learned, calibrated, or theoretically optimal weighting and not an algorithmic performance contribution. Its formal invariants are: an unsafe or non-executable candidate is never selected; required named-state coverage is complete before its threshold is evaluated; gold-derived correctness is absent from the decision record; equal scores resolve by the frozen order; and an empty eligible set yields abstention. These invariants support conformance testing and replay determinism. They do not imply semantic correctness, calibrated confidence, or superiority over a simpler policy.

Question-derived shape checks are deliberately weak. They detect cues such as “how many”, “highest”, “first”, and explicit limits and compare them with the projected aggregate or ordering structure. They do not infer business semantics such as whether “active” refers to an asset, alarm, or work-order state. Lexical value hits likewise reward only presented values traceable to the question and context; they are not evidence that the associated column is the intended one. These low-bandwidth features keep adjudication inspectable but explain why large tie sets remain possible.

Abstention has two layers. Pre-ranking abstention occurs when no candidate passes safety and execution requirements or when required state coverage is incomplete. Post-ranking ambiguity is currently resolved by stable order rather than abstention because no pre-registered margin threshold exists in v3. The high tie rates observed later show that future work should freeze an ambiguity policy—for example, abstaining on unresolved top ties or requesting a user clarification—before examining new outcomes.

Robustness is treated as a vector. Tested dimensions are database mutation denial, bounded execution, complete named-state eligibility, and three constructed metamorphic witnesses. Candidate-order stability is explicitly not established (101 versus 117–118/180 with 130/180 top ties), and frozen-reference agreement supplies neither qualified semantic gold nor operational correctness. Figure 6 later summarizes these claim ceilings.

Version attribution is non-retroactive: release v3 supplies the 5760 attempts and 80/100/101 counts under its read-only, authorizer, opcode/time, and row limits; the 14-test FINAL executor supplies only later control-conformance evidence; formal-v5 supplies the 25,920-row state diagnostic; and superseded R3 remains review provenance without a retained scientific count. Algorithm 1 states the resulting gold-isolated coordination rule.

Algorithm 1 Gold-isolated deterministic coordination

Require: question x_i , schema S_i , external candidates $\mathcal{Y}_i$, read-only executor, named state evidence

- 1: post structured intent and schema grounding to the append-only blackboard
 - 2: canonicalize and deduplicate externally produced candidates
 - 3: **for** each candidate $y \in \mathcal{Y}_i$ **do**
 - 4: apply single-statement read-only validation
 - 5: record execution and available reference-free state evidence
 - 6: **end for**
 - 7: **if** required named-state coverage is incomplete for a candidate **then**
 - 8: mark that candidate ineligible (fail closed)
 - 9: **end if**
 - 10: **if** no safe executable candidate meets all registered gates **then**
 - 11: abstain
 - 12: **else**
 - 13: apply the fixed validation, state-evidence, coverage, and order rule
 - 14: **end if**
 - 15: post the decision and seal the blackboard
 - 16: load reference results only for offline evaluation
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3.5. Inherited GridDB Factorial Design

The paired design crosses context package $c \in \{0, 1\}$ and composite hint $h \in \{0, 1\}$. F00 uses full DDL and global values without the hint; F01 adds the hint; F10 uses compact domain-grounded context without the hint; and F11 combines compact context and the hint. The same 180 questions appear in all four cells for each backbone, yielding 720 Qwen and 720 Granite predictions.

Because each question appears in every prompt cell, cell differences are paired within question. The two backbone snapshots form parallel finite-corpus experiments rather than independent replications drawn from a model population. Difficulty labels and normalized-SQL clusters are retained as descriptive structure. Cluster-based randomization reduces the influence of near-duplicate SQL forms, while composition-sensitivity intervals show how estimates vary when observed clusters are resampled. Neither procedure converts this development-visible synthetic set into a population sample.

Qwen2.5-Coder-7B Q4_K_M and Granite-3.3-8B Q4_K_M were run at temperature zero with fixed snapshots. Each prompt produced exactly one response. The parser retained the first SQL candidate through the first semicolon; the validator then applied the read-only policy. There was no candidate ranking, multi-agent negotiation, or repair in this experiment.

Strict execution equality is primary. A common-target projected-column indicator is a secondary manipulation check. Paired finite-set contrasts use normalized-SQL clusters as dependence proxies, registered sign-flip tests, composition-sensitivity bootstrap intervals,

and Holm correction. The intervals describe sensitivity to the observed corpus composition, not population confidence.

3.6. Component, Multi-State, and Public Protocols

The separately frozen component study contains 700 scored calls. E1 compares candidate generation with and without presented value evidence on 170 paired questions in 61 frozen groups. E2 requests up to three candidates and compares a deterministic reference-answer-independent selector with the first candidate on 180 questions in 70 groups. Question-level paired differences are averaged with equal question weights; bootstrap resampling and sign assignments operate at group level. Selection is sealed before gold loading. Rescue, harm, and oracle-at-three are descriptive; oracle-at-three is a gold-only upper-bound diagnostic. The three two-member Holm families are E1 across backbones, E2 across backbones, and the two cross-backbone modifiers.

The v5 reliability protocol reproduced all 1440 T0 labels and then executed every prediction over 18 states. The primary 66-question analysis uses a question-weighted logical-AND endpoint over 15 semantic states. A prediction passes only when it agrees with the reference on every included state. The nine registered contrasts form a separate Holm family; with only 12 structural groups, all $2^{12} = 4096$ group-sign assignments are enumerated. The sharp-null interpretation assumes exchangeability of those 12 group signs; the small group count limits randomization resolution, and the groups are dependence proxies rather than sampled deployment clusters.

The BIRD protocol compares direct prompting, decomposition, schema selection, and bounded execution repair. Each method has a fixed call pattern and zero retries. Accuracy and method differences retain equal item weighting over 500 items; pairwise randomization assigns one common sign to all paired item differences within each of 11 databases and applies Holm correction across 12 comparisons. Its sharp-null interpretation assumes database-level sign exchangeability. Eleven clusters give coarse attainable evidence and do not represent a random sample of power-grid databases.

The four protocols answer different questions and are not pooled. GridDB factorial cells test prompt-package manipulations on a controlled grid schema. E1 and E2 test presented values and reference-free selection under a prospectively frozen call-and-selection procedure on development-visible items. Formal-v5 tests agreement across constructed states. BIRD tests non-grid workflow portability. A positive result in one protocol cannot substitute for a missing end-to-end multi-agent comparison in another.

For every inferential family, multiplicity is defined before interpretation. The primary GridDB factorial family, the corrected projected-column family, the three component families, the nine multi-state contrasts, and the BIRD pairings remain separate because their endpoints and experimental units differ. Exact enumeration is used when the registered cluster count permits it; otherwise the retained composition-sensitivity procedure is reported. Effect sizes and complete counts remain visible even when a multiplicity-adjusted rule is not met.

3.7. Retrospective Offline Coordination Diagnostic

The replay verifies SHA-256 for two 720-row prediction ledgers and the 25,920-row formal-v5 ledger before reading them. For each question it pools the eight existing outputs from two backbones and four prompt cells, canonicalizes SQL, and removes duplicates. Frozen T0 fields provide execution and metadata-shape evidence. The Critic records named-state evidence as unavailable because existing state-agreement labels are gold-relative.

This is a coverage diagnostic. Candidates were generated under different models and prompt packages; the pool is not a deployable, matched multi-agent run. No selected output is compared with gold to produce an accuracy number.

3.8. Deterministic Descriptive Offline Re-Execution Study v3

The release-v3 study reuses exactly eight frozen source slots for each of the 180 GridDB evaluation questions: four Qwen prompt cells followed by four Granite prompt cells. It performs no LLM call and creates no new question or SQL label. Before any method selects, the three methods share one precomputed eight-by-four execution ledger: T0 plus three metamorphic witnesses for every slot. Thus 5760 is the shared physical evidence-collection count, not a separate natural runtime cost assigned to each selector. The fixed-order control ignores the collected ranking evidence and chooses slot 0. Validation-only ranks candidates without named-state evidence, whereas complete-witness selection requires all three reference-free metamorphic outcomes. Incomplete coverage or no passing candidate causes abstention.

The three query-blind witnesses have distinct SHA-256 values and are built from T0 without reading questions, prompts, predictions, scores, or gold. M1 adds an irrelevant relation and rows; M2 adds harmless indexes and rebuilds storage; M3 adds nullable probe columns to three business tables. M1 and M2 preserve original-schema denotation. M3 preserves explicitly projected original columns but deliberately allows wildcard projection to fail. Query order is preserved when candidate SQL contains ORDER BY; otherwise rows are compared as a multiset. The selector sees only question text extracted from frozen prompt ledgers, the schema, eight SQL strings, executor traces, and T0-to-witness equivalence. It does not see gold SQL, answer-shape metadata, correctness, difficulty, feature tags, required literals, or gold-relative formal-v5 labels.

These witnesses test metamorphic relations rather than alternative ground-truth answers. M1 asks whether an irrelevant table changes behavior, M2 whether logically neutral storage changes alter the result, and M3 whether projection is stable when nullable columns are appended. A pass means that the observed result satisfies the registered relation; it does not mean the original query expresses the user's intent. A fail can reveal brittleness, as with wildcard projection, but may also reflect a transformation outside the intended query equivalence class. For this reason, witness outcomes enter eligibility only under the disclosed construction and remain reference-free diagnostics.

Candidate order is a first-class experimental variable because the adjudicator uses stable order for equal scores. Original order places four Qwen slots before four Granite slots and follows the prompt-cell sequence within each backbone. Reverse-order sensitivity changes only this tie breaker after the same evidence has been collected. It is therefore a direct diagnostic of unresolved ranking information, not an alternative model or a new sample. The large change later observed under reversal motivates an abstention or clarification policy for future work.

Before v3, 21 artifacts were bound under SHA-256 8b34bc370451173197ad07b460908537836409d6eb49f303caf46d13d53889e6; 21 is a manifest-entry count. The restored tree passes 30 tests (10 agent, 9 release-v3, 4 replay, and 7 executor). A superseded 10-test R3 suite lacked raw-BLOB accounting, and an invalid transient 33/33 run after overwriting the frozen executor was reverted and excluded. Each v3 run sealed all 180 boards before directly loading raw gold and retained 1440 source slots and 5760 attempts without retry; v3a and v3b outputs were byte-identical. Audit confirmed mechanical traceability but found that pre-run tests had opened same-item v2 gold-derived rows. The controlling SUPERSESSION_NOTICE.json therefore classifies v3 as descriptive re-execution; v1 and v2 remain excluded predecessors.

Descriptive endpoints are strict execution accuracy over all 180 questions, coverage and abstention, complete-three-witness invariance of the selected candidate, and rescue/harm relative to fixed order. The retained sensitivity matrix crosses three weight policies, invariant-pass thresholds of 1/3, 2/3, and 3/3 while always requiring complete observation, and original versus reversed candidate-order tie breaking. Because same-item outcomes were accessible before v3, this matrix is reported only as descriptive analysis.

The later FINAL executor is a post-review engineering correction and is not part of the historical v3 test count. Its 14-test suite closes the raw-BLOB accounting defect found during R3 review, while the immutable R3 implementation and reviews remain preserved outside this FINAL candidate. No v3 result was rerun or reinterpreted with the corrected boundary.

3.9. Generative-AI Assistance and Author Responsibility

The local SQL-generation experiments used the explicitly identified Qwen2.5-Coder-7B-Instruct Q4_K_M and Granite-3.3-8B-Instruct Q4_K_M snapshots under the frozen protocols described above. Separately, during reconstruction and revision of this manuscript, the authors used the OpenAI Codex agent interface to propose and edit prose, suggest experimental and audit checks, review and draft Python/L^AT_EX code, assemble provenance and response matrices, and assist local build and visual-quality checks. The exact backend model identifier and version for every Codex session were not retained in the project records; this provenance boundary is reported explicitly, and no unrecorded model identifier is inferred.

All executable suggestions were run locally against retained inputs, and reported numerical results were taken from frozen or independently recomputed ledgers rather than accepted from generated prose. The authors retained responsibility for protocol authorization, evidence-class decisions, interpretation, citations, declarations, and the final manuscript. Codex output was not treated as qualified power-system expert annotation or adjudication. External machine-generated question-SQL candidates remain labeled machine silver and do not become human gold through model review. All six scientific figures are deterministic code-native graphics with file-level lineage; no generative-image model supplied quantitative evidence.

4. Results

No result in this section estimates the benefit of the five-role architecture relative to a call-matched single-agent or monolithic controller. RQ1 is answered only by software conformance: typed traces, fail-closed coverage, read-only enforcement, deterministic replay, and retained failures behaved as specified in the reported tests. The remaining subsections report inherited prompt/component experiments and outcome-exposed historical-pool diagnostics under their separate estimands.

4.1. Complete GridDB Cell Results

All 1440 scheduled predictions are present, and independent SQLite re-execution reproduced the stored execution verdicts. Table 4 reports strict execution equality, and Figure 2 visualizes the same audited cell estimates.

Table 4. Strict execution equality over 180 GridDB questions per cell.

Backbone	F00 full/no hint	F01 full/hint	F10 compact/no hint	F11 compact/hint
Qwen	76/180 (0.4222)	129/180 (0.7167)	78/180 (0.4333)	108/180 (0.6000)
Granite	77/180 (0.4278)	100/180 (0.5556)	74/180 (0.4111)	108/180 (0.6000)

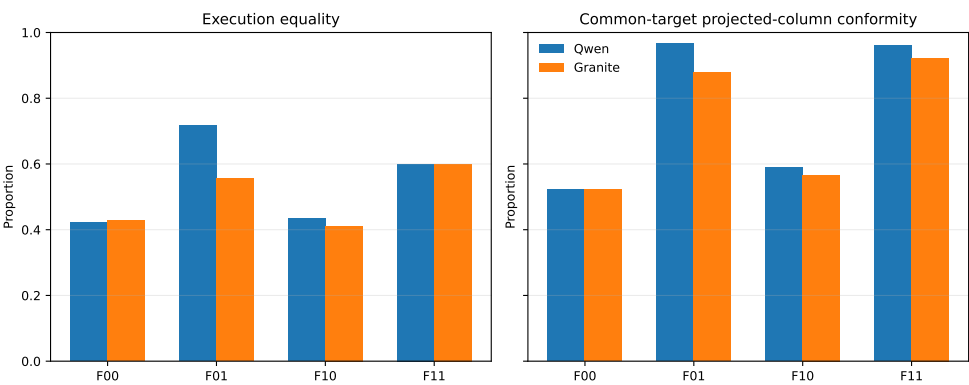


Figure 2. Audited GridDB cell point estimates. Each cell contains all 180 attempts; the figure is descriptive and does not replace registered factorial contrasts.

The common-target projected-column diagnostic is 0.5222 in F00 for both backbones. Qwen obtains 0.9667, 0.5889, and 0.9611 in F01, F10, and F11; Granite obtains 0.8778, 0.5667, and 0.9222. Returning the requested number of columns is therefore easier than returning the intended rows. In Qwen F01, projected-column conformity is 0.9667 while execution equality is 0.7167.

The eight cells also show why a single best prompt cannot represent the design. Adding the hint to full context increases the point estimate for both backbones, whereas adding compact context without the hint leaves execution equality close to F00. Under compact context, the hint produces 108 correct answers for both models, but this numerical convergence arises from different unhinted baselines and does not establish identical behavior. Complete cell reporting preserves these interactions rather than selecting only the highest cell.

The 0.2500 gap between Qwen F01 projected-column conformity and strict execution equality warns against structural proxy endpoints. A model may recognize that one column, an aggregate, or an ordered row is requested while still choosing the wrong table, join, predicate, value, or boundary. Shape diagnostics can localize errors and inform adjudication, but they cannot replace result equality or qualified semantic review.

4.2. Factorial Effects and Multiplicity

Table 5 reports the complete primary family. For Qwen, the composite-hint execution effect is +0.2306, the package–hint interaction is −0.1278, and the compact-package main effect is −0.0528. Granite’s corresponding estimates are +0.1583, +0.0611, and +0.0139. Zero of nine primary execution tests survives Holm correction. Qwen’s hint effect has raw/adjusted $p = 0.01372/0.09604$, its interaction 0.00919/0.07352, and the cross-backbone hint modifier 0.00640/0.05760. No primary factorial execution effect or modifier is therefore described as statistically nonzero.

The normalized-SQL mapping contains 70 groups, 58 singletons, and a maximum group size of 19. Question-level weighting is retained; groups are resampling and sign-assignment proxies, not verified authoring templates or sampled clusters. The randomization interpretation requires exchangeability of group signs under the tested sharp null, while the composition interval asks how the finite-corpus mean changes when those observed groups are reweighted. Neither assumption follows from the data, and the high singleton fraction limits protection against unmeasured dependence.

The displayed composition intervals are pointwise sensitivity intervals; they are neither simultaneous/familywise intervals nor inversions of the nine Holm-adjusted randomization tests. An interval that excludes zero can therefore coexist with a Holm non-rejection.

Table 5. Complete primary factorial family for strict execution equality. Estimates are question-weighted finite-set differences; intervals are 95% normalized-SQL-group composition-sensitivity intervals. Each sign-randomization used 70 groups, 100,000 draws, and the displayed raw values were Holm-adjusted jointly over all nine rows.

Scope	Effect	Estimate	Composition interval	Raw p	Holm p
Qwen	Compact-package main	-0.0528	[-0.1131, 0.0000]	0.10523	0.52614
Qwen	Structural-hint main	0.2306	[0.0548, 0.4355]	0.01372	0.09604
Qwen	Package $\times$ hint	-0.1278	[-0.2454, -0.0341]	0.00919	0.07352
Granite	Compact-package main	0.0139	[-0.0584, 0.0798]	0.85021	1.00000
Granite	Structural-hint main	0.1583	[-0.0193, 0.3662]	0.15480	0.61919
Granite	Package $\times$ hint	0.0611	[-0.0539, 0.2013]	0.55890	1.00000
Granite-Qwen	Backbone $\times$ package	0.0667	[-0.0360, 0.1814]	0.41478	1.00000
Granite-Qwen	Backbone $\times$ hint	-0.0722	[-0.1229, -0.0276]	0.00640	0.05760
Granite-Qwen	Three-way interaction	0.1889	[0.0090, 0.4321]	0.07010	0.42060

The multiplicity decision, not interval exclusion, controls statements about the registered nine-test family.

No minimum practically important execution-accuracy difference, equivalence margin, or non-inferiority margin was registered. Therefore the study cannot infer practical equivalence from a non-rejection, and the 1/180 complete-witness difference is only a mechanism trace rather than a meaningful-benefit result.

In the separately corrected secondary family, the common-target hint effects survive: +0.4083 for Qwen (adjusted $p = 0.000090$) and +0.3556 for Granite (adjusted $p = 0.01944$). Because the hint supplies structural information later scored by this endpoint, these are manipulation-check results, not independent semantic evidence.

The multiplicity result changes the emphasis of the factorial study. Large raw contrasts motivate hypotheses about structural instructions, but none of the nine primary execution comparisons meets the registered familywise rule. The projected-column findings verify that the manipulation changed a closely aligned surface property. They do not show that the compact package or hint produces a backbone-independent accuracy gain. Future work should separate schema pruning, value presentation, normalization, and structural hints rather than retaining composite packages.

4.3. Component Results

The component run completed all 700 scored calls under a prospectively frozen call-and-selection procedure on development-visible GridDB items. On the paired 170-question subset, Qwen first-candidate correctness changed from 83/170 to 101/170, a +0.1059 difference (95% composition-sensitivity interval [+0.0282, +0.2013], adjusted $p = 0.0310$). Granite remained 69/170 in both paired conditions, with an interval spanning zero. The cross-backbone modifier did not meet the registered rule, so the Qwen result is not replicated across backbones and is not a five-role result.

The deterministic selector changed 24 Qwen and 36 Granite choices. Relative to the first candidate, it rescued eight and harmed one Qwen question, and rescued ten while harming none for Granite. Execution effects were +0.0389 and +0.0556; neither survived its two-test Holm family. Oracle-at-three values of 0.6389 and 0.4667 are gold-only diagnostics and are not deployable selector results.

For Qwen, the value-evidence intervention adds 18 correct first candidates when the 170 eligible V0 questions are compared with their V1 counterparts under the registered analysis; the corresponding Granite estimate is zero. The disagreement is more informative than averaging the backbones: the effect depends on how a frozen model uses schema and value context. Oracle-at-three measures candidate-pool headroom, whereas the reference-

free selector measures how much of that headroom the recorded rule can recover without gold.

Complete component counts are retained in the numerical supplement. On all 180 V1 questions, Qwen first/selector/oracle-at-three counts were 105/112/115, and Granite counts were 71/81/84; the paired 170-question intervention counts and all registered effects appear in Table 6. These are component, not five-role, outcomes.

Table 6. Complete component-effect families. Intervals are pointwise group-composition sensitivity summaries, not simultaneous intervals and not inversions of the Holm tests. Each family contains the two displayed members indicated by its identifier.

Family	Contrast	N	Groups	Estimate	Composition interval	Raw <i>p</i>	Holm <i>p</i>
E1	Qwen V1–V0 first candidate	170	61	0.1059	[0.0282, 0.2013]	0.01552	0.03104
E1	Granite V1–V0 first candidate	170	61	0.0000	[-0.1902, 0.1705]	1.00000	1.00000
E2	Qwen selector–first V1	180	70	0.0389	[-0.0081, 0.1071]	0.50080	0.50080
E2	Granite selector–first V1	180	70	0.0556	[0.0075, 0.1232]	0.06425	0.12850
Cross-E	Granite–Qwen E1 effects	170	61	-0.1059	[-0.2701, 0.0394]	0.26986	0.53971
Cross-E	Granite–Qwen E2 effects	180	70	0.0167	[-0.0062, 0.0457]	0.37477	0.53971

The three retained two-member Holm families are E1 across backbones, E2 across backbones, and Cross-E across modifiers. The Granite E2 pointwise interval excludes zero while its adjusted decision does not meet the registered rule; these summaries are not contradictory. The 61- and 70-group mappings retain question weights but are dependence proxies rather than a probability sample. Figure 3 visualizes these effects without replacing the adjusted decisions.

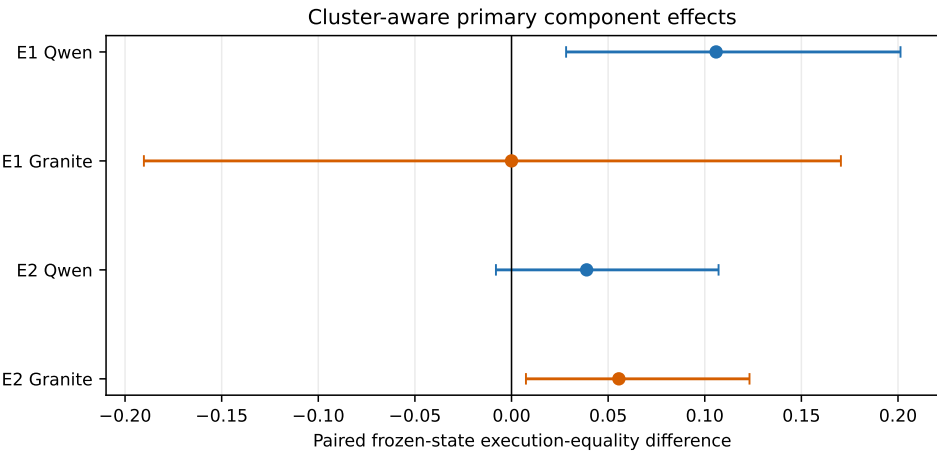


Figure 3. Component effects under a prospectively frozen call-and-selection procedure on development-visible GridDB items. Intervals describe composition sensitivity; the registered decision also requires multiplicity-adjusted evidence.

4.4. Multi-State Reliability Results

The v5 scorer produced all 25,920 registered rows, and its T0 slice reproduced 1440/1440 snapshot labels. Table 7 and Figure 4 report the 66-question automatic subset: 15-state logical-AND rates range from 0.6212 (Granite F01) to 0.8182 (Granite F00). All nine Holm-adjusted values equal 1.0000 under exact cluster-sign enumeration. No context, hint, interaction, or cross-backbone multi-state effect meets the declared rule. These rates characterize constructed-state agreement, not population accuracy or human-certified equivalence.

The ordering under this endpoint differs from the single-snapshot table. Granite F00 has the highest logical-AND rate on the 66-question automatic subset even though Granite F11 has the larger T0 execution count over all 180 questions. The denominators and

endpoints differ, so this is not a contradiction. It demonstrates why a robustness statement must name both the admitted question subset and the required state set.

Table 7. Complete 15-state logical-AND counts for the prediction-blinded 66-question automatic subset.

Backbone	F00	F01	F10	F11
Qwen	49/66 (0.7424)	48/66 (0.7273)	53/66 (0.8030)	43/66 (0.6515)
Granite	54/66 (0.8182)	41/66 (0.6212)	51/66 (0.7727)	49/66 (0.7424)

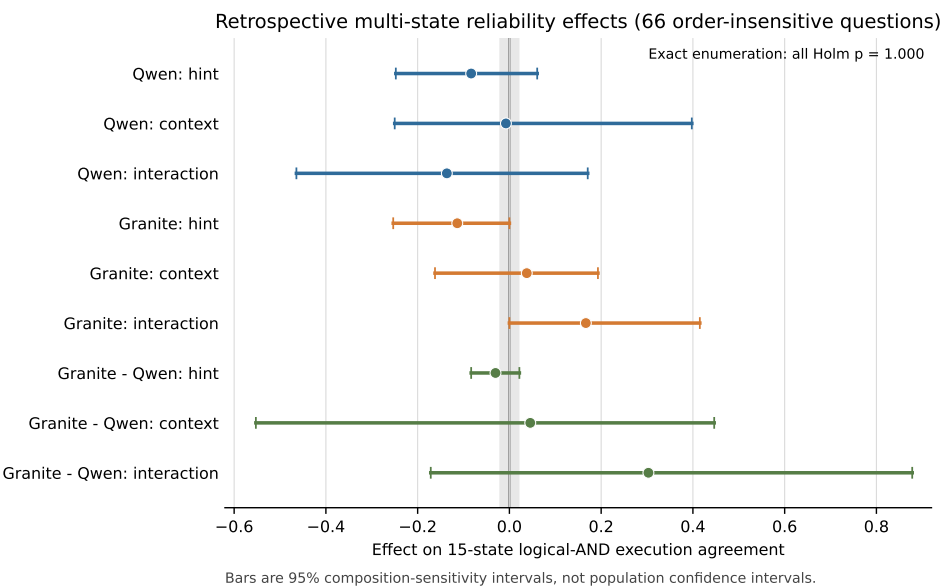


Figure 4. GridDB multi-state logical-AND agreement for the prediction-blinded automatic subset. The endpoint is a constructed-state robustness diagnostic.

4.5. Public BIRD Results

Table 8 reports all eight cells. BIRD Mini-Dev is a public non-grid portability benchmark and does not validate power-grid semantics. B0–B2 use one physical generation call per item; B3 uses two, so the inherited comparison estimates frozen workflow efficacy under unequal call counts rather than a call-matched repair effect. Each backbone completed 2500 calls, 2000 final predictions, zero retries, and zero final-ledger omissions; latency and token totals were not promoted into the retained summary and are therefore reported as unavailable rather than reconstructed.

After Holm adjustment, Qwen decomposition is 0.076 below direct prompting (adjusted $p = 0.0430$), and schema selection is 0.092 above decomposition (adjusted $p = 0.0117$). No Granite contrast survives correction. Different method orders across the two model snapshots argue against a universal prompting recipe. The two failed incident runs totaling 2476 physical calls remain excluded and immutable; they are not retries contributing to these denominators.

The BIRD results also constrain the meaning of “agentic” improvement. Decomposition reduces Qwen accuracy relative to direct prompting, while bounded repair uses twice the physical calls without becoming the best Qwen method. Granite shows a different descriptive ordering and no adjusted contrast. More stages or calls therefore do not guarantee a better outcome. A future comparison must count physical generations, retries, context tokens, and failed attempts, not only final predictions.

Table 8. Inherited BIRD Mini-Dev v1.1 results (non-grid portability evidence; 11 database clusters). Intervals are database-cluster composition-sensitivity intervals.

Backbone	Method	Correct/500	Accuracy	Interval	Calls/item	Final- ledger omissions
Qwen	B0 direct	189/500	0.378	[0.286, 0.470]	1	0
Qwen	B1 decomposition	151/500	0.302	[0.203, 0.401]	1	0
Qwen	B2 schema selection	197/500	0.394	[0.279, 0.511]	1	0
Qwen	B3 execution repair	174/500	0.348	[0.250, 0.452]	2	0
Granite	B0 direct	102/500	0.204	[0.125, 0.284]	1	0
Granite	B1 decomposition	105/500	0.210	[0.131, 0.294]	1	0
Granite	B2 schema selection	101/500	0.202	[0.137, 0.273]	1	0
Granite	B3 execution repair	118/500	0.236	[0.159, 0.318]	2	0

4.6. Retrospective Replay Coverage

The replay audited all 180 GridDB questions without a generation call. At least two distinct frozen SQL candidates were present for 173 questions. After safety checks and binding to consistent T0 execution evidence, 172 questions had at least two eligible candidates and entered retrospective adjudication. Seven failed closed because only one unique SQL remained, and one failed because fewer than two candidates were eligible.

Of 180 audited questions, 173 had at least two unique frozen SQL candidates and 172 retained at least two eligible candidates for retrospective adjudication; seven failed closed with one unique candidate and one with fewer than two eligible candidates. No reference-free named-state evidence was available. Unique-candidate multiplicities from one through eight occurred for 7, 19, 15, 18, 27, 23, 45, and 26 questions, respectively. These counts establish interface coverage, not correctness, execution gain, rescue, or multi-agent superiority.

4.7. Descriptive Offline Coordination-Selection Results

The deterministic descriptive re-execution retained 5760 candidate-state attempts; 332 were non-executable or authorization/resource/SQL failures and remained in the all-attempt ledger. Every question nevertheless had at least one eligible candidate, so both adjudicating selectors covered 180/180 questions and abstained on 0/180. This zero-abstention result is a consequence of at least one eligible slot plus forced stable-order resolution of post-ranking ties, not confidence-calibrated coverage. Table 9 reports the recorded rules.

Keeping all 332 failures matters for two reasons. Removing them would overstate the executability of the historical pool, and retaining their slot identities verifies that selectors did not receive replacements or retries. Full question-level coverage means only that another slot was eligible for every question; it does not mean every candidate was valid or that an operational system should always answer.

Table 9. Descriptive release-v3 re-execution over one historical eight-slot pool per question. All methods choose after the same precomputed candidate/evidence ledger of 32 attempts per question. Accuracy is finite-corpus strict execution equality after all blackboards were sealed; this is not a new-generation or five-role end-to-end experiment.

Selector	Correct/180	Accuracy	Covered	Abstained	Invariant	Rescue/harm
Fixed order, shared ledger	80/180	0.4444	180	0	177/180	0/0
Validation rank, shared ledger, no CF	100/180	0.5556	180	0	179/180	22/2
Complete-three-witness selector	101/180	0.5611	180	0	180/180	23/2

Figure 5 juxtaposes the two recorded endpoints. Validation-only and complete-witness selection differ by one frozen-reference match and one invariant selection. This small incremental change contrasts with their 20–21-match difference from fixed order and focuses interpretation on validation ranking and tie policy rather than on a broad counterfactual benefit.

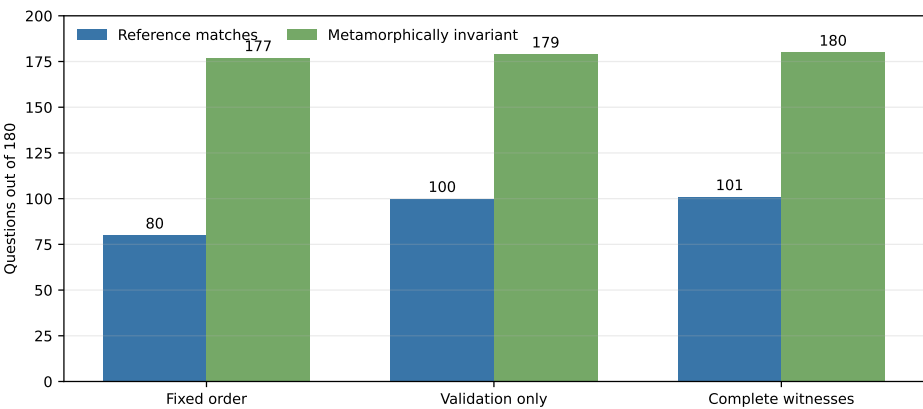


Figure 5. Descriptive release-v3 selector endpoints over 180 historical-pool questions. Reference matches are loaded only after board sealing; invariance denotes agreement across the three constructed witnesses. Bars are not a prospective multi-agent comparison.

The two adjudicating selectors each had multiple highest-scoring candidates on 130/180 questions. Mean top-tie sizes were 5.4000 and 5.3889; all eight candidates tied on 79 validation-only and 78 complete-witness questions. With original-order tie breaking, 178/180 selections came from Qwen slots, reflecting source order rather than learned preference.

Thus, most apparent source preference is mechanically induced. When all eight candidates tie, the first slot is selected regardless of backbone quality; when seven tie, the earliest surviving slot has the same advantage. The observed 178 Qwen selections cannot be interpreted as an adjudicator comparison between Qwen and Granite. It instead measures how often the available reference-free features failed to separate candidates before the stable-order rule acted.

The complete multiplicity and selected-slot distributions are retained in the numerical supplement. Validation-only tie-size counts for sizes 1–8 were 50/5/3/4/5/8/26/79; complete-witness counts were 50/5/3/4/5/9/26/78.

Validation-only ranking added 20 net correct items relative to fixed order through 22 rescues and 2 harms. Complete-witness selection changed the chosen candidate on only Q039, yielding one further frozen-reference match (101/180 versus 100/180), one additional rescue, no additional harm, and no coverage or abstention change. Table 10 gives the complete constructed trace.

M3 added nullable columns to `work_orders`; consequently, complete-witness eligibility rejected the wildcard projection and retained the explicit `work_order_id`, `scheduled_date` projection. Because neither query resolves the request’s status and date-boundary ambiguity through qualified review, this observation is not called an engineering correctness rescue. It is a synthetic, outcome-exposed projection-stability trace, not evidence of general counterfactual reasoning, semantic validity, robustness, coordination, or five-role benefit.

Table 11 reports all 18 sensitivity cells. Every original-order cell produced 101/180; reverse order produced 117/180 under default and intent-emphasis weights and 118/180 under validation-equal weights. These more favorable values are outcome-exposed diag-

Table 10. Q039 is the sole validation-only versus complete-witness selection difference. Correctness is agreement with the frozen synthetic reference, not an independently adjudicated engineering judgment.

Field	Validation-only	Complete witnesses	Interpretation
Natural-language request	“Show the first scheduled work order after June 2024.”		Ambiguous: “after June” can imply after 30 June, and “scheduled” can denote a status or merely a scheduled date.
Selected query	SELECT * FROM work_orders WHERE scheduled_date > '2024-06-01' ORDER BY scheduled_date ASC LIMIT 1	SELECT work_order_id, scheduled_date FROM work_orders WHERE scheduled_date > '2024-06-01' ORDER BY scheduled_date ASC LIMIT 1	Both omit an explicit scheduled-status predicate and use a date boundary whose intended meaning lacks expert adjudication.
M3 outcome	Fails when nullable columns extend wildcard width	Preserves the explicit two-column projection	Constructed projection-stability behavior
Frozen-reference match	Incorrect	Correct	Outcome-exposed 1/180 trace only

nostics, not preferred alternatives. The 16–17-item shift and high tie multiplicity show that the apparent selector behavior is materially order-dependent.

Threshold invariance across 1/3, 2/3, and 3/3 occurs because the selected candidates’ recorded witness patterns and score ties do not change the final original-order choices under these policies. It does not show that threshold selection is generally unimportant. Conversely, the reverse-order increase cannot justify choosing reverse order after seeing gold. Together, the cells diagnose inadequate discriminative evidence and define a pre-registration requirement for the next pool.

Table 11. Complete-witness selector sensitivity summarized across invariant-pass thresholds. Results were identical at 1/3, 2/3, and 3/3 for every weight/order pair; coverage was 180/180 throughout.

Weight policy	Thresholds represented	Original order correct/180	Reverse order correct/180
Default	1/3, 2/3, 3/3	101 (0.5611)	117 (0.6500)
Intent emphasis	1/3, 2/3, 3/3	101 (0.5611)	117 (0.6500)
Validation equal	1/3, 2/3, 3/3	101 (0.5611)	118 (0.6556)

The full machine-generated numerical supplement binds all eight GridDB cells, 12 component endpoints, eight 15-state cells, 18 sensitivity cells, the 360-row item-level tie ledger, selected-origin counts, Q039 trace, source paths, hashes, and byte counts in R3_EVIDENCE_TABLES.json and companion CSV files. Table 8’s final column records only omissions from the accepted final ledgers; it is not a taxonomy of parsing, authorization, execution, provider, or incident failures. The two excluded BIRD incidents are reported separately above. This result supports auditability and exposes a design risk; it does not establish a generally robust or superior selector.

Across the results, the strongest positive numbers occur under different evidence classes: Qwen F01 is the best GridDB generation cell, schema selection is the best Qwen BIRD workflow, complete-witness original-order selection yields 101 historical-pool matches, and reverse order raises the descriptive count to 117–118. These values cannot be combined into one framework accuracy because they use different questions, candidates, calls, visibility histories, and selectors. Their joint value is diagnostic: they identify components for a clean prospective protocol and claims that remain unsupported.

5. Discussion

5.1. What the Evidence Supports

The combined evidence supports four bounded findings. First, explicit structural and SQL-operation instructions strongly alter projected-column adherence, but surface conformity does not establish correct execution. Second, the 700-call component experiment, whose call-and-selection procedure was prospectively frozen on development-visible items, finds that presented value evidence matters for one frozen Qwen snapshot while showing no effect for Granite; this is component evidence, not a five-role outcome. Third, candidate selection and prompting are backbone-dependent: the component selector did not meet its registered efficacy rule, and the descriptively best BIRD method differed between Qwen and Granite. Fourth, on the historical eight-slot pool, validation ranking changed the observed finite-set score relative to fixed order, but complete witness evidence changed only Q039, and top-score ties on 130/180 items made results strongly dependent on arbitrary source order.

These findings justify separating intent, schema grounding, synthesis, validation, and state evidence. They do not establish that five roles are superior to one call or that newly generated multi-agent candidates would reproduce the offline gain. The architecture’s demonstrated value is methodological and safety-bounded: typed contracts, database-enforced eligibility, complete-evidence gates, deterministic selection, abstention, incident retention, and a sealed gold boundary make later comparisons reproducible and falsifiable.

Figure 6 maps each evidence source to its claim ceiling. The map prevents results from being promoted merely because they appear in the same implementation: executor tests establish a software boundary, GridDB and component studies establish finite-corpus behavior, BIRD establishes non-grid portability, and v3 establishes descriptive selection on a previously evaluated pool. Prospective five-role benefit and operational validity remain untested.

Evidence classes and claim ceiling	
Executor and trace tests	implemented safety boundary
GridDB/component protocols	finite-corpus component evidence
BIRD Mini-Dev	non-grid portability evidence
Historical-pool re-execution	descriptive selector behavior
Prospective five-role benefit	not evaluated
Operational grid validity	not evaluated

Figure 6. Evidence classes and permitted claim ceiling. Colors distinguish audited or bounded evidence, non-grid portability, descriptive prior-outcome-exposed analysis, and evaluations not performed.

5.2. Interpreting the Retrospective Replay

The replay asks whether inherited ledgers contain enough distinct executable candidates to exercise the coordination interfaces. For 172 of 180 questions, they do. This reduces implementation risk and identifies eight fail-closed cases that require a registered fallback or a new matched generation policy.

The replay cannot estimate a coordination effect. Its pool mixes two backbones and four prompt packages, so generation condition and diversity are confounded. It also lacks reference-free counterfactual equivalence evidence. Post hoc gold scoring of selected outputs would remain exploratory and cannot substitute for a prospective matched comparison.

5.3. Interpreting the Descriptive Re-Execution

The release-v3 study hash-binds the selection view, rules, code, tests, and state inputs and verifies the software fact that each execution path directly loaded raw gold only after all blackboards were sealed. This does not establish outcome blindness: its frozen test suite had accessed gold-derived v2 evaluation rows for the same items before v3. The additive supersession notice is the first authority for interpreting the immutable release package and invalidates its historical prospective, prespecified, and outcome-unseen labels. The study is therefore a deterministic descriptive re-execution over a historical candidate pool. Each selector inherited the same precomputed eight-slot candidate/evidence ledger; this does not mean equal historical model calls between source conditions, and it does not turn those candidates into contemporaneous agent outputs.

The small contrast between validation-only and complete-witness selection is a mechanism trace, not a gain estimate. V3 uses three different query-blind database hashes and exercises an irrelevant-relation witness, an index/storage witness, and a nullable schema-extension witness. These are constructed operator families rather than independent samples from a deployment population. The full gate changes only Q039 because M3 exposes wildcard-projection arity while preserving explicit-column projections. The natural-language request itself admits unresolved status and date-boundary interpretations, so the frozen-reference match cannot be promoted to power-grid semantic correctness. An earlier v1 diagnostic used three same-hash insertion files and is retained as an excluded incident; v2 is retained as a provenance-incomplete predecessor. The supersession notice excludes their prospective interpretations and none of these predecessors supports the current scientific claim.

The tie audit quantifies a second limitation: 130/180 questions have multiple highest-scoring candidates for each adjudicating selector, with approximately 5.4 candidates in the average top-tie set. The original-order rule consequently chooses source slots without semantic evidence and selects Qwen-origin candidates on 178/180 items. Reporting only the 0.5556 default or choosing the 0.6556 reverse-order sensitivity after outcome access would be misleading. A later rule-development study requires development-only calibration, richer reference-free evidence, and a genuinely untouched evaluation resource.

5.4. Power-Grid Use and Reproducibility

Database-enforced read-only execution, resource limits, syntactic validity, and result-shape conformity are necessary safeguards, not operational certification. A safely executed query can still return the wrong assets, omit time bounds, misinterpret codes, or mishandle ties. Table/column/function allowlists constrain SQL operations inside this local evaluator; user authorization requires a separately authenticated identity, institutional access policy, and data-owner decision. The additive FINAL raw-cell-byte, budgeted-result-byte, and output-width controls were not used in release v3 and cannot be invoked to reinterpret its results. They bound returned values within one Python/SQLite process, not SQLite's entire temporary-memory or process-memory footprint. Multi-state witnesses reduce accidental snapshot equality but cannot cover all distributions or schema changes. Generated SQL and returned source rows require authorized human inspection before any maintenance, protection, dispatch, or asset-management decision.

Negative results are part of the contribution. Zero of nine primary factorial execution tests survives correction. The deterministic component selector meets its efficacy rule for neither snapshot. No registered multi-state effect survives correction. Granite and Qwen do not share the same best BIRD method. Immutable ledgers, direct re-execution, registered test families, portable manifests, and explicit incidents preserve these outcomes rather than collapsing them into a success narrative.

6. Limitations and Future Work

The primary domain corpus is one synthetic database with eight tables, 98 rows, and 180 evaluation questions. Its evaluation partition was visible during development. The 70 structural groups are dependence proxies, and 58 are singletons. Results cannot be generalized to production schemas, permissions, missing-data patterns, units, access-control policies, or workloads.

Only two quantized local snapshots were evaluated. Model family, size, quantization, serving software, and serialization may interact. The compact package bundles schema reduction, values, normalization, and serialization; the hint bundles projection, aggregation, grouping, ordering, and limit rules. The inherited factorial experiment identifies package effects rather than contributions of individual internal rules.

The five-role core and frozen release-v3 executor pass 30 unit/adversarial/regression tests; the separate FINAL executor passes 14 additive tests, including four new BLOB-accounting and trace/boundary methods. The 21-entry freeze manifest is not a test count, and the invalid transient 33/33 run is excluded. The system has not completed a new-generation experiment on an evaluation resource whose outcomes were previously inaccessible to the study process. Its Analyst and Cartographer are deterministic skeletons, its Synthesizer packages external candidates, and its adjudication weights have not been independently optimized. Release v3 descriptively re-executes selection over historical candidates and previously evaluated items; it does not estimate generation, communication, latency, token, or autonomous-agent effects. Existing formal-v5 agreement labels are gold-relative and intentionally excluded from selection.

RTS-GMLC, SimBench, and NERC question-SQL assets remain machine-adjudicated silver data. Qualified review is required for intended projections, units, ordering, tie handling, and result granularity before external accuracy is reported.

Future work should execute a new hash-locked, call-matched comparison of four contemporaneous generation conditions. These conditions are a single direct candidate, staged question/schema handoffs with one candidate, a fixed multi-candidate budget with validation and adjudication but without named-state evidence, and the same candidate pool with a pre-registered multi-family reference-free state suite. Model snapshot, question order, database state, decoding, candidate count, physical calls, and evaluator must be held constant. The protocol must pre-register execution accuracy, valid SQL, abstention, tokens, latency, failure taxonomy, clustered inference, multiplicity, and incident retention. The candidate-order rule must be chosen on development-only evidence before the untouched evaluation set is opened.

Further work can replace lexical grounding with a trained schema linker, add a low-confidence full-schema fallback, compare deterministic adjudication with budget-matched voting, and evaluate risk-coverage calibration. Qualified power-system/database reviewers must independently assess domain semantics; LLM or author-assisted triage cannot be relabeled as independent expert gold. Every change should receive a new protocol identity rather than modifying a completed run.

7. Conclusions

MA-SQLGrid reframes power-grid text-to-SQL as an auditable coordination problem. Five typed software roles separate question analysis, schema mapping, candidate packaging, safety and execution validation, and named-state criticism. A deterministic controller records their handoffs, selects only eligible candidates, and abstains when required evidence is incomplete. Here, *multi-agent* denotes this inspectable role decomposition, not demonstrated benefit from five autonomous language-model agents. The frozen executor enforces read-only and resource limits at the SQLite boundary rather than relying on lexical filtering alone; the separate FINAL variant adds tested raw-cell-byte, budgeted-result-byte, width, and function controls that are not retroactive to v3 and are not whole-process memory isolation.

The inherited experiments provide a substantial but bounded evidence base. The 1440-prediction GridDB study shows strong projected-column instruction uptake but no primary execution effect surviving Holm correction. The 700-call component study supports presented values for one Qwen snapshot, not Granite, and does not validate its selector under the registered rule. The 25,920-row state study finds no corrected factorial agreement effect. The 5000-call BIRD protocol supplies a non-grid public comparison whose best method depends on the backbone and whose methods use unequal call counts. The retrospective replay confirms candidate coverage for 172 of 180 questions without accuracy claims.

The descriptive offline re-execution records 80 of 180 matches (44.4%) for fixed order, 100 of 180 (55.6%) for validation-only, and 101 of 180 (56.1%) for complete witnesses. Both adjudicating selectors have highest-score ties on 130 of 180 questions (72.2%), with an average tie multiplicity of approximately 5.4. Reversing candidate order changes the complete-witness sensitivity result to 117–118 of 180. The sole Q039 difference is a constructed, outcome-exposed, semantically ambiguous projection trace. These facts, the historical mixed-source pool, development-visible GridDB, and absent qualified semantic review prevent a general multi-agent superiority or power-grid deployment claim. The original title is retained as the framework identity; *robust* refers only to explicitly tested mutation-denial, bounded-execution, complete-evidence, and three-witness mechanisms, not universal or operational robustness.

Supplementary Materials: The FINAL editor/reviewer package is generated from an explicit al-lowlist and clean-extraction checked. It contains the exact manuscript source and PDF, bibliography, all six figure artifacts and lineage sources, complete numerical tables, the 360-row tie ledger, Q039 trace, supersession notice, corrected FINAL executor and tests, review records, the itemized response matrix, rights inventory, audit reports, and hashes. It excludes superseded manuscripts and PDFs, credentials, caches, and excluded incident outputs presented as evidence. Repository publication, third-party permissions, qualified expert review, and untouched end-to-end evaluation remain separate requirements.

Author Contributions: Conceptualization, B.L. and Y.Y.; methodology, B.L.; software, B.L.; validation, B.L. and C.S.; formal analysis, B.L. and C.S.; investigation, B.L. and C.S.; resources, Y.Y.; data curation, B.L. and C.S.; writing—original draft preparation, B.L.; writing—review and editing, C.S. and Y.Y.; visualization, B.L.; supervision, Y.Y.; project administration, Y.Y.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Science and Technology Research Project of State Grid Fujian Electric Power Co., Ltd., grant number 521300250006. The funder had no role in the study design; in the collection, analysis, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Institutional Review Board Statement: Not applicable. This study did not involve human participants or animals.

Informed Consent Statement: Not applicable.

Data Availability Statement: The implementation repository is available at <https://github.com/gaoxingkele/ma-sqlgrid>, and BIRD Mini-Dev remains available from its public provider. The manuscript-bound rights-safe verification package, including the numerical supplement, code, tests, rights inventory, supersession notice, and independent audits, is available from the corresponding author upon reasonable request. Owing to third-party licensing restrictions, restricted source records and source-dependent RTS-GMLC or SimBench artifacts may be shared for editorial and peer-review verification only when permitted by the applicable third-party licenses. Reviewer access does not constitute a redistribution license.

Acknowledgments: During reconstruction and revision of this manuscript, the authors used the OpenAI Codex agent interface for prose drafting and editing, experimental and audit suggestions, Python/L^AT_EX code review and drafting, provenance assembly, reproducibility checks, and local build/visual-quality assistance. The exact backend model identifier/version was not retained in the project records; no unrecorded identifier is inferred. All reported values were obtained from retained executable ledgers and local analyses; Codex output was not treated as qualified expert annotation or independent scientific adjudication. All six scientific figures are code-native and no generative-image output is used as quantitative evidence. The authors reviewed and edited the outputs and take full responsibility for the publication.

Conflicts of Interest: The authors declare no conflicts of interest.

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